INTRODUCTION

PROJECT TITLE: CashLite – A Community Savings and Financial Management Application

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My name is Mas'ud Al-Hassan (KASU/DIP/CSE/24/0013), presenting my project titled CashLite: A Community Savings and Financial Management Application.

CashLite is a modern, cross-platform mobile application engineered using the Flutter framework and Dart programming language. The platform was developed to digitize, organize, and streamline core financial operations within community savings groups by consolidating member administration, contribution tracking, pool management, and micro-loan workflows into a centralized digital system.

PROBLEM STATEMENT

Traditional community savings groups and cooperative networks heavily rely on manual, paper-based record-keeping methods.

This legacy approach introduces critical system vulnerabilities and operational friction:

1. **Arithmetic & Record-Keeping Errors:** Manual ledgers regularly lead to calculation mistakes, ledger discrepancies, and unverified contribution entries.
2. **Tracking Challenges:** Monitoring individual member contributions over extended cycles and tracking outstanding loan repayments across manual logs is difficult and error-prone.
3. **Lack of Financial Transparency:** Opaque record-keeping breeds mistrust among participants regarding pool liquidity and rotational payout schedules.
4. **Administrative Inefficiency:** Managing group finances manually is excessively time-consuming, unscalable, and susceptible to physical document loss.

These operational bottlenecks established the necessity for an organized, reliable, and accessible software solution to automate community financial administration.

PROPOSED SOLUTION

CashLite directly resolves these manual inefficiencies by implementing an integrated digital ledger system.

The application unifies essential financial activities within a single, reactive environment:

1. **Centralized Financial Workspace:** Replaces physical logbooks with a digital repository for real-time record verification.

2. **Automated Ledger Management:** Simplifies contribution entry, savings monitoring, and rotational payout organization.

3. **Structured Micro-Loan Processing:** Provides clear oversight of loan applications, tracking them from submission through approval and repayment statuses.

4. **Accountability & Transparency:** Standardizes group data to maintain record integrity and foster trust among all participants.

SYSTEM ARCHITECTURE & KEY FUNCTIONAL DOMAINS

The CashLite mobile application is architected across five core functional modules:

1. **Home Domain:** Serves as the primary executive dashboard, displaying high-level liquidity metrics, savings summaries, and recent transaction updates.

2. **Members Domain:** Manages member profiles, role assignments, rotational payout schedules, and active contribution statuses.

3. **Savings Domain:** Provides dedicated functionality for logging, verifying, and auditing individual member contributions.

4. **Loans Domain:** Manages the entire micro-loan lifecycle, allowing administrators to evaluate applications, track approval states, and monitor repayments.

5. **Profile Domain:** Manages user account settings, session parameters, and system details.

Together, these interconnected modules provide a comprehensive framework covering the operational scope of community financial groups.

TECHNICAL IMPLEMENTATION

CashLite was engineered using Flutter and Dart to leverage single-codebase cross-platform compilation and dynamic user interface rendering.

Current Engineering Deliverables

1. Responsive UI layout built according to modern mobile usability principles.

2 . Modular software design separating UI view components from application state data

THE FIVE MAIN PAGES OF THE CASHLITE APP

1. Home: Displays the main dashboard and a summary of the group's financial activities.
2. Members: Shows and manages the members of the savings group.
3. Saving: Allows users to record and monitor their savings contributions.
4. Loans: Allows users to manage loan applications, approvals, and repayments.
5. Profile: Displays and manages the user's account information and settings.

IDEAS FOR FUTURE DEVELOPMENT

To scale CashLite for commercial deployment, subsequent development will focus on:

1. Database Integration: Connecting persistent backends (e.g., SQLite/Firebase/Hive) for permanent data storage.

2. Authentication Security: Implementing secure user login gateways and role-based permissions (RBAC).

3. Advanced Financial Analytics: Generating downloadable PDF reports and automated accounting balance sheets.

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CashLite

SAVE TOGETHER. GROW TOGETHER.

Welcome back

Enter your details to continue.

Full name

Password

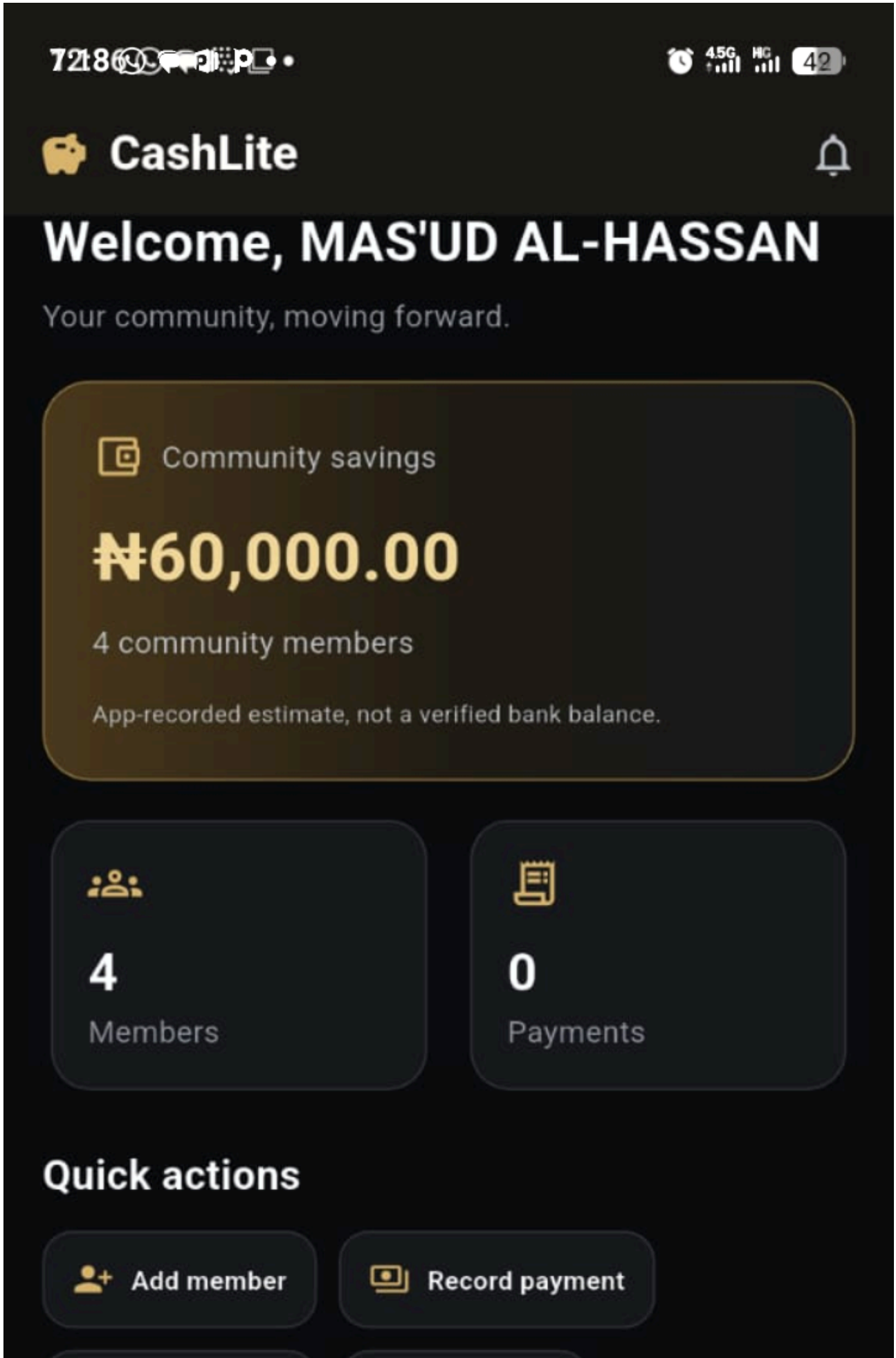


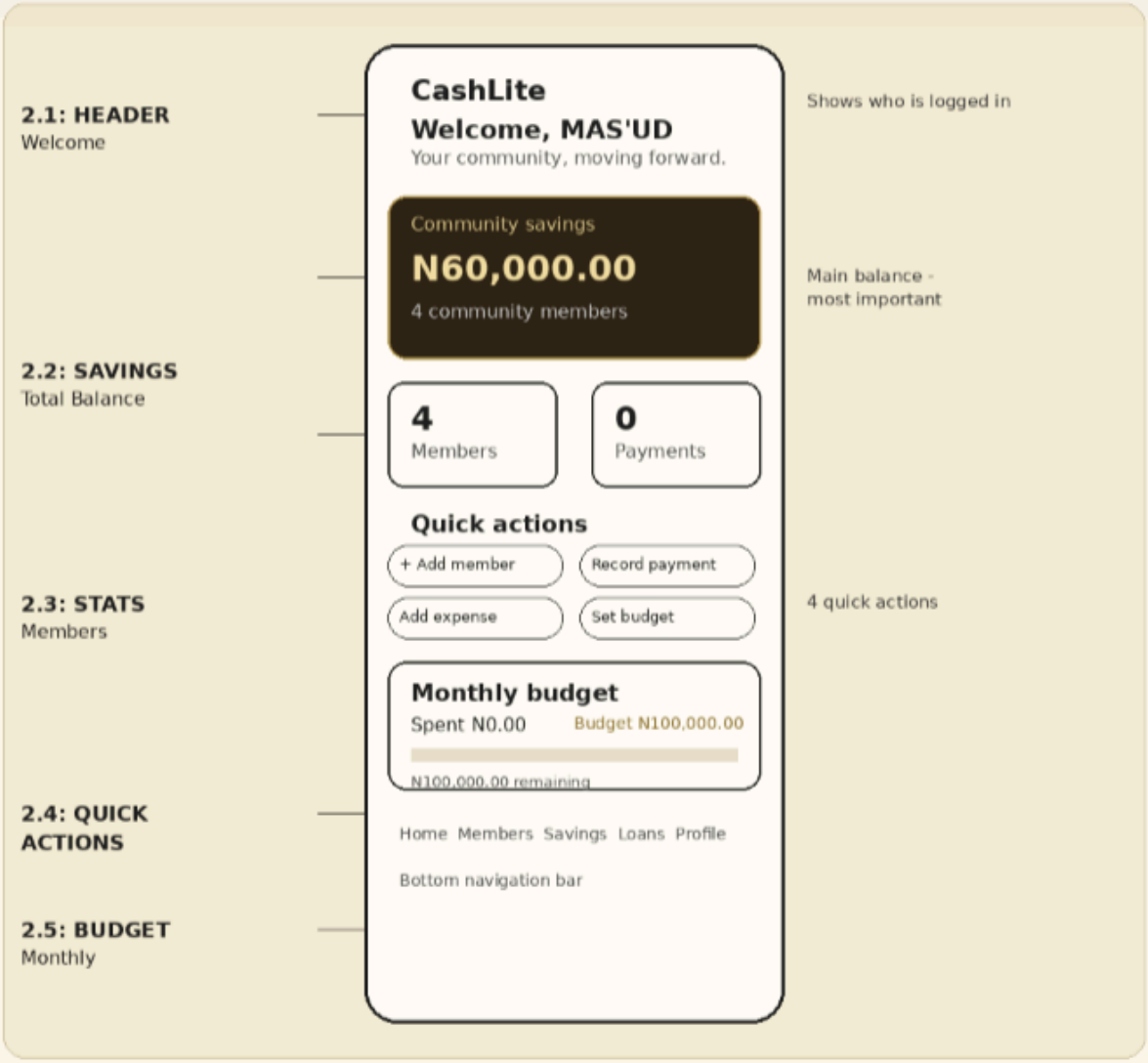
1. Explanation

Figure 1: Welcome Back Screen - Sketch

This is the first screen of CashLite. It shows the CashLite logo and asks the user to enter Full Name and Password. Tapping Continue opens the Home Page. The eye icon lets the user show or hide the password, and the bottom text welcomes the user to the app.

The screen is simple and clear. It helps the user log in quickly and start using the app without confusion.


CashLite
Welcome, MAS'UD AL-HASSAN
Your community, moving forward.
Community savings
₦60,000.00
4 community members
App-recorded estimate, not a verified bank balance.
4
Members
0
Payments
Quick actions
Add member
Record payment



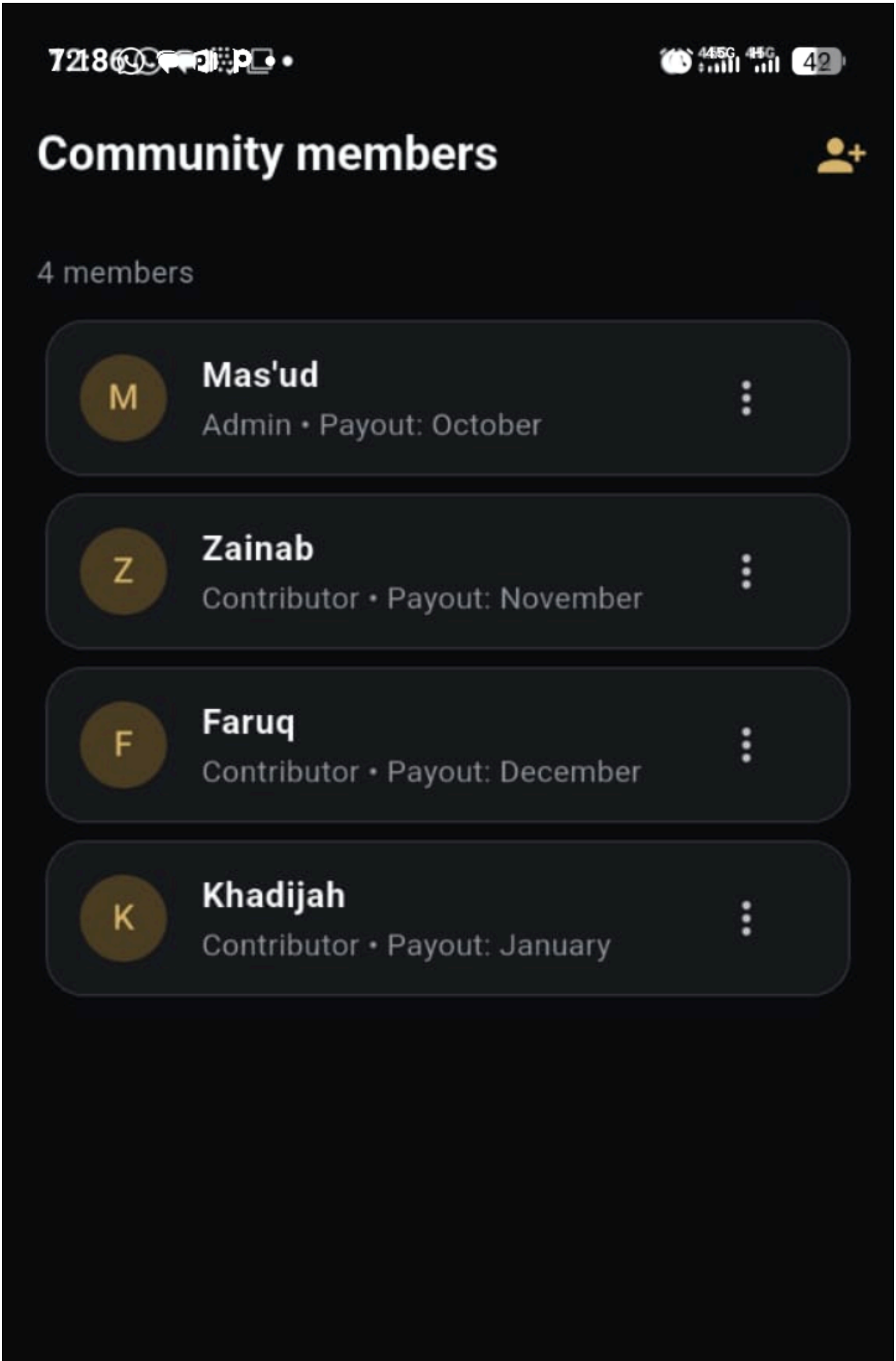
2. Explanation

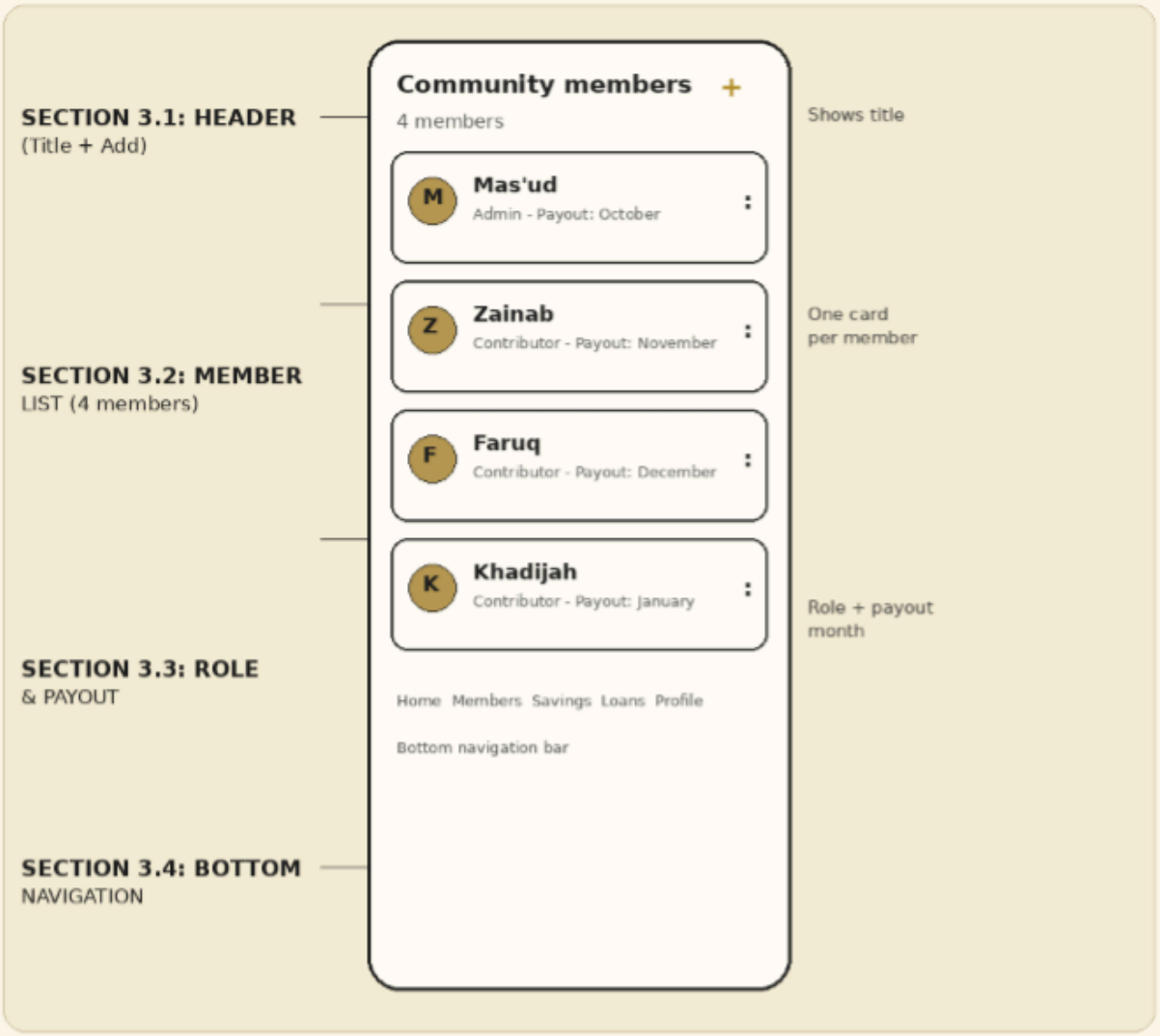
Figure 2: Home Page - Sketch

This is the Home Page of CashLite. It welcomes the user by name - MAS'UD AL-HASSAN - and shows a message, "Your community, moving forward."

The gold Community Savings card shows the total savings of N60,000.00 for 4 community members. Below it, the Members and Payments cards give a quick summary.

The Quick Actions buttons let the user Add Member, Record Payment, Add Expense and Set Budget quickly. The Monthly Budget card shows Spent, Budget and Remaining. The bottom bar lets the user move to Home, Members, Savings, Loans and Profile.


Community members
4 members
M
Mas'ud
Admin • Payout: October
Z
Zainab
Contributor • Payout: November
F
Faruq
Contributor • Payout: December
K
Khadijah
Contributor • Payout: January



3. Explanation

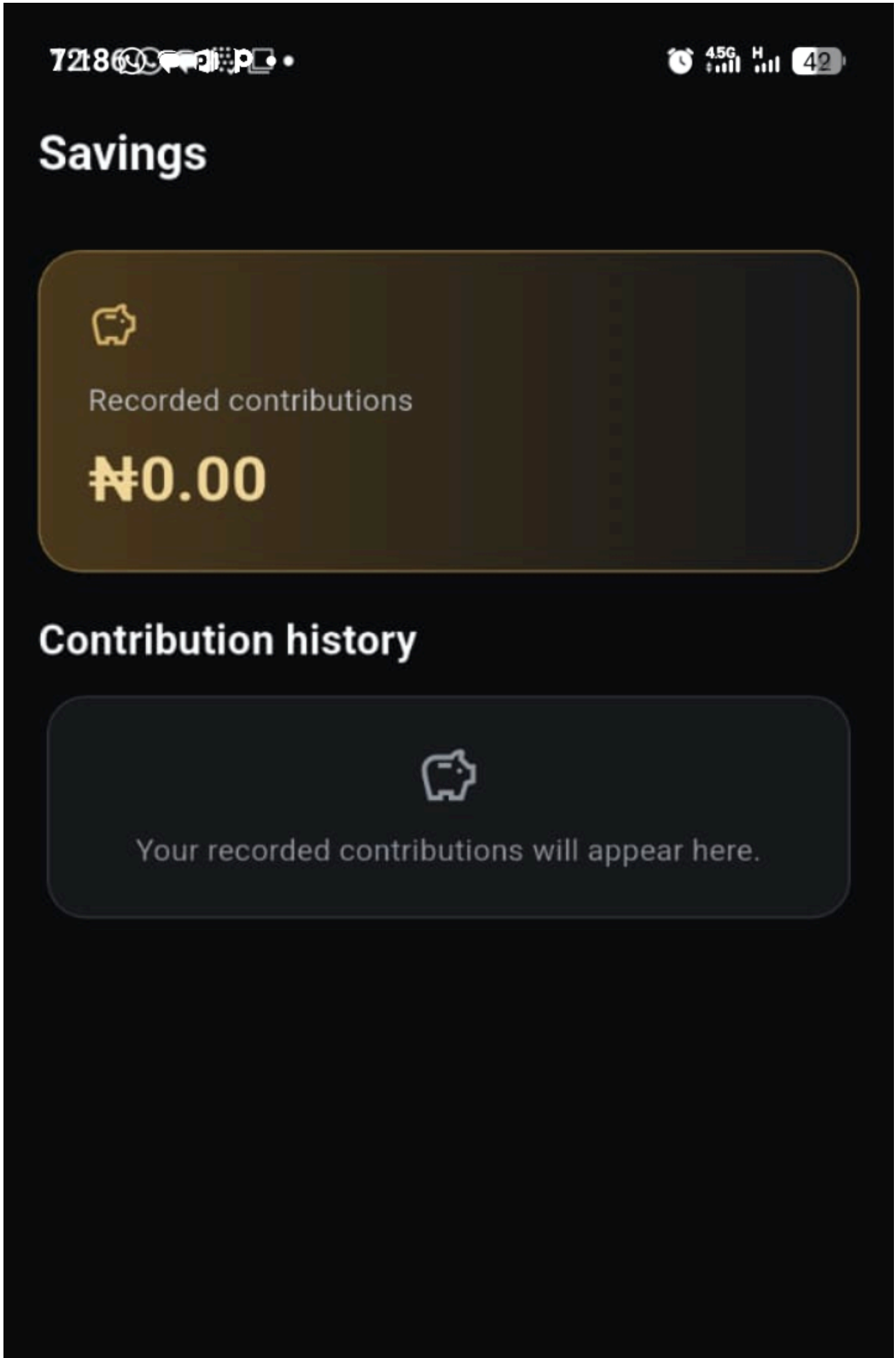
Figure 3: Community Members Page - Sketch

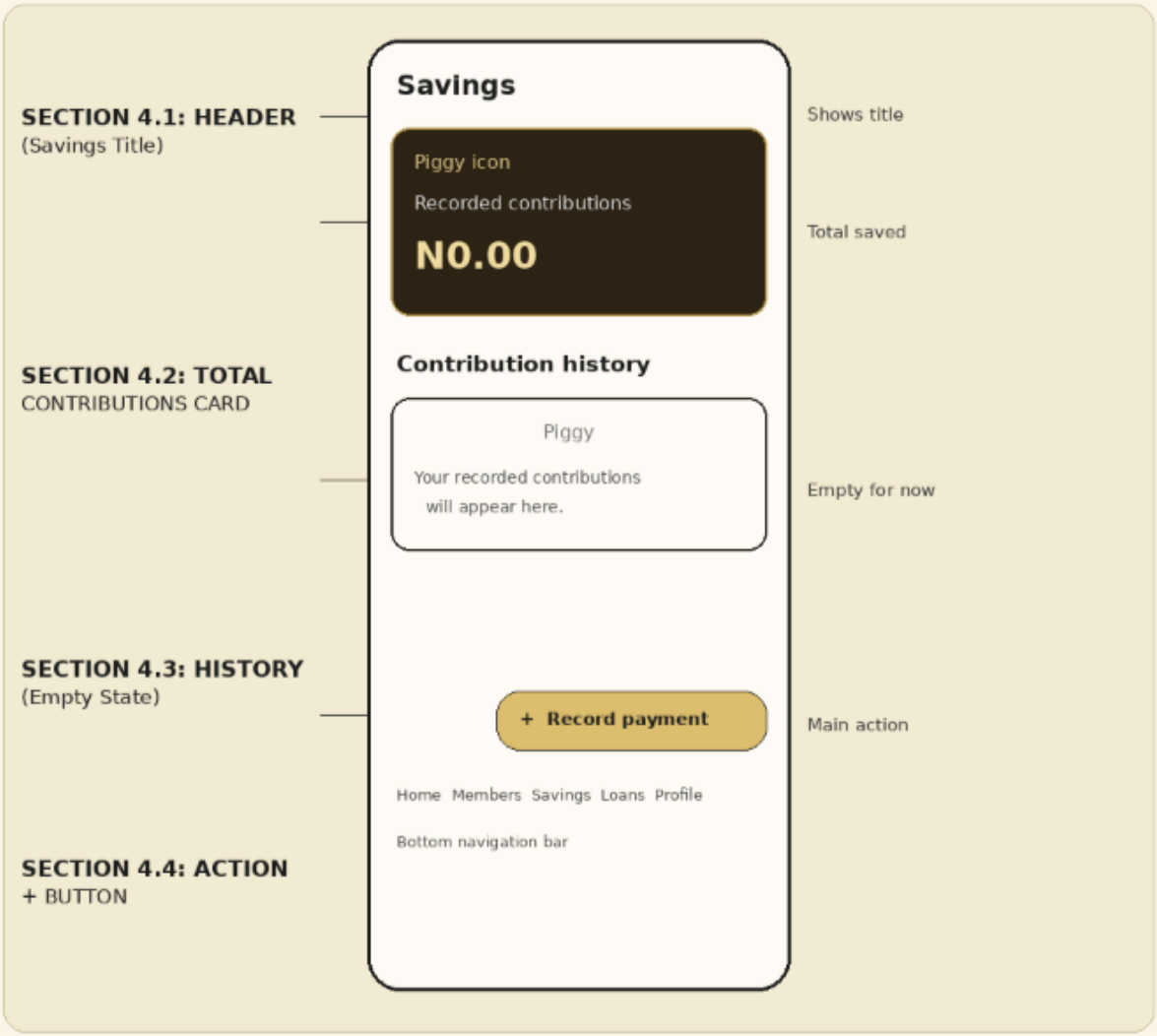
This is the Community Members Page of CashLite. It shows all the people in the savings group.

At the top, it shows the title "Community members" and "4 members". The + icon at the top right lets the admin add a new member.

Each member has a card with their initial, name, role and payout month. For example, Mas'ud is the Admin with payout in October, Zainab in November, Faruq in December and Khadijah in January.

The three dots on each card let the user view or manage that


Savings
Recorded contributions
₦0.00
Contribution history
Your recorded contributions will appear here.



4. Explanation

Figure 4: Savings Page - Sketch

This is the Savings Page of CashLite. It shows how much the group has contributed.

At the top, it shows the title "Savings". The gold card shows "Recorded contributions" with a total of N0.00. This means no contribution has been recorded yet.

Below it, the "Contribution history" section is empty and says "Your recorded contributions will appear here." When a payment is recorded, it will appear here.

The gold "+ Record payment" button at the bottom right lets

Community loans

Outstanding approved loans

₦50,000.00

Yusuf Audu — Approved

₦50,000.00

Equipment purchase

Repaid: ₦0.00

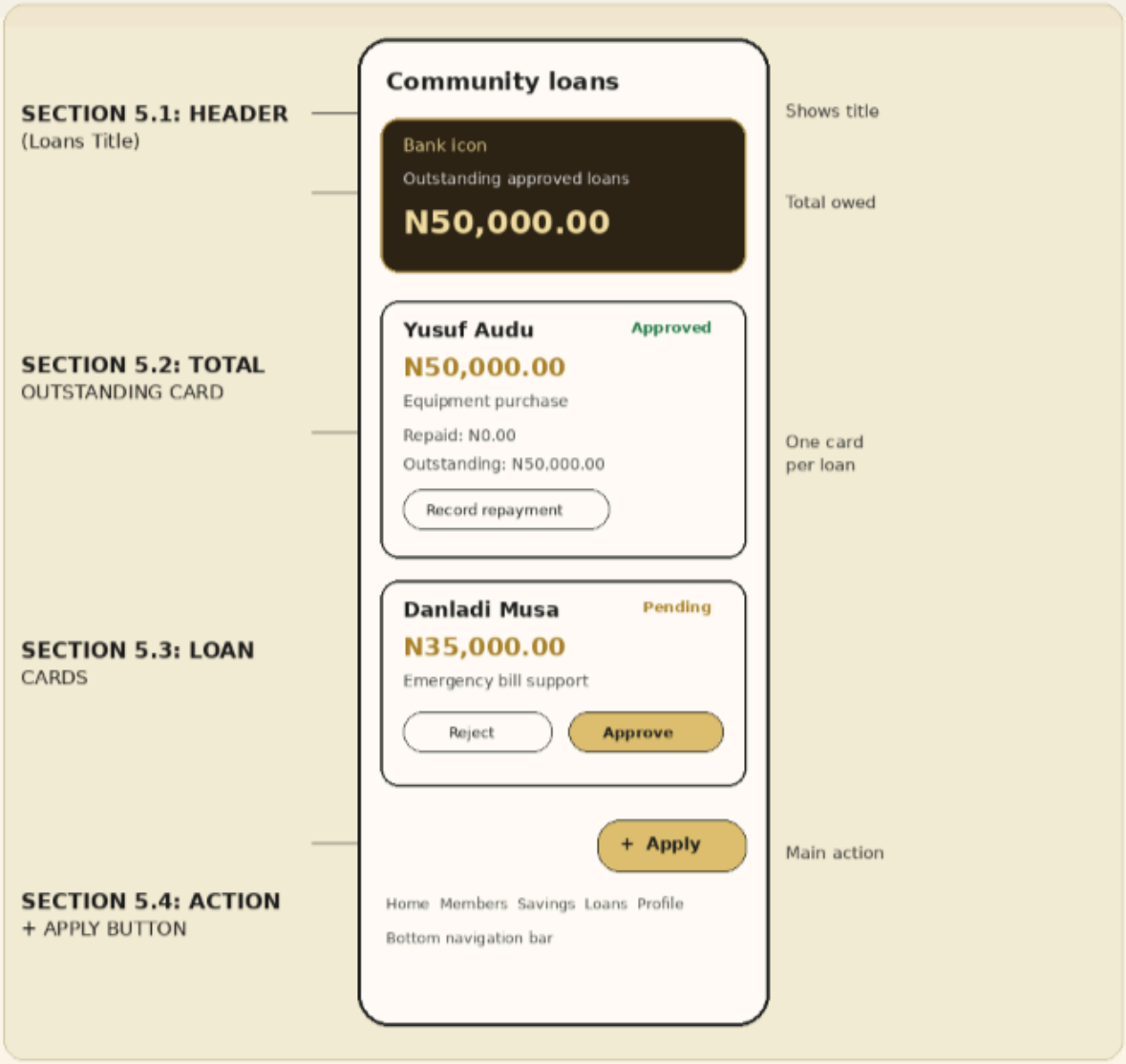
Outstanding: ₦50,000.00

Record repayment

Danladi Musa — Pending

₦35,000.00

Emergency bill support



5. Explanation

Figure 5: Community Loans Page - Sketch

This is the Community Loans Page of CashLite. It shows all the loans in the group.

At the top, it shows the title "Community loans". The gold card shows "Outstanding approved loans" with a total of N50,000.00.

Each loan has a card with name, amount, purpose and status. For example, Yusuf Audu borrowed N50,000.00 for Equipment purchase. It is Approved. He has repaid N0.00 and still owes N50,000.00. Danladi Musa asked for N35,000.00 for Emergency bill support. It is Pending.

The buttons let the admin Record Repayment, Reject or Approve

My profile



MAS'UD AL-HASSAN

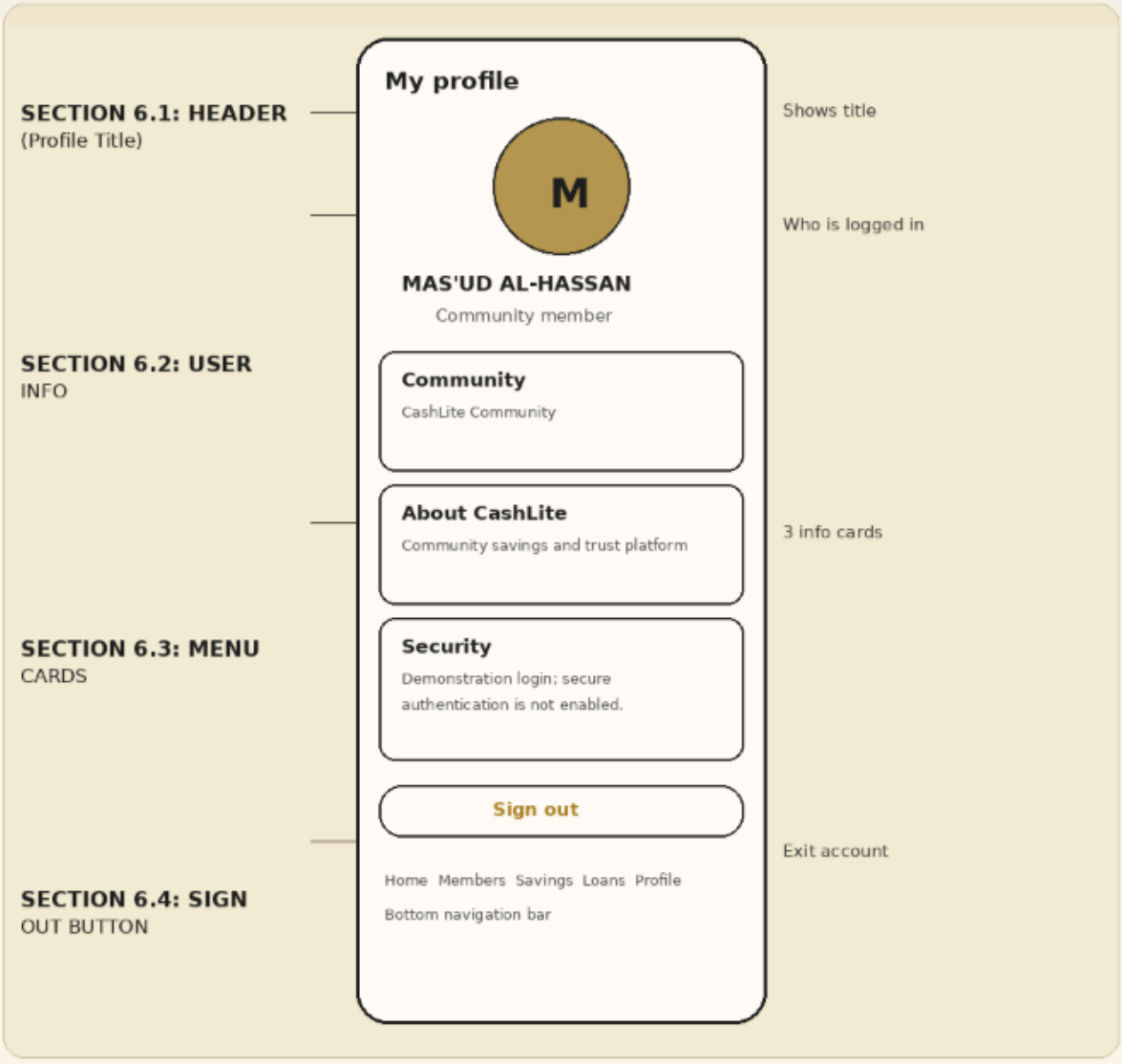
Community member

Community
CashLite Community

About CashLite
Community savings and trust platform

Security
Demonstration login; secure authentication is not enabled.

Sign out



6. Explanation

Figure 6: My Profile Page - Sketch

This is the My Profile Page of CashLite. It shows who is logged in and the app information.

At the top, it shows the title "My profile". Below it is the user's picture, name MAS'UD AL-HASSAN and role "Community member".

The three cards show: Community - CashLite Community, About CashLite - Community savings and trust platform, and Security - Demonstration login; secure authentication is not enabled.

The "Sign out" button at the bottom lets the user log out

CONCLUSION

CashLite demonstrates a practical application of modern mobile software engineering to solve real-world financial management challenges within community savings groups.

By digitizing member records, contribution histories, and loan workflows, the project bridges the gap between traditional manual cooperative practices and modern fintech systems. It highlights core computer science competencies in UI design, system structuring, and mobile application architecture.

GITHUB REPOSITORY

https://github.com/masudalhassan82/Cashlite-school-project

LIVE WEB VERSION

https://masudalhassan82.github.io/Cashlite-school-project